INTERFACE OF SMART GARBAGE SYSTEM IN GATED COMMUNITIES

A Mini Project Report

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CERTIFICATE

This is to certify that the project work entitled "INTERFACE OF SMART GARBAGE SYSTEM IN GATED COMMUNITIES"

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students of Electronics and Communication Engineering Department, Vasavi College of Engineering in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering is a record of the Bonafide work carried out by them during the academic year 2025-2026. The result embodied in this project report has not been submitted to any other university or institute for the award of any degree.

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DECLARATION

This is to state that the work presented in this thesis titled "INTERFACE OF SMART GARBAGE SYSTEM IN GATED COMMUNITIES" is a record of work done by us in the Department of Electronics and Communication Engineering, Vasavi College of Engineering, Hyderabad. No part of the thesis is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the project work done entirely by us and not copied from any other source. We hereby declare that the matter embedded in this thesis has not been submitted by us in full or partial thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

Improper waste management in gated communities often leads to overflow of garbage bins, unhygienic surroundings, foul smell, and health-related issues. A major challenge in conventional systems is the inability to segregate waste at source and monitor dry and wet bin levels independently. This project presents an Interface of Smart Garbage System designed for gated communities using embedded systems and IoT technology.

In the proposed system, an IR sensor detects when waste is placed at the bin inlet and triggers the processing cycle. A moisture sensor then classifies the waste as wet or dry. Based on this classification, a servo motor physically rotates to direct the waste into the appropriate compartment — the wet waste section or the dry waste section. Two HC-SR04 ultrasonic sensors independently measure the fill levels of the dry and wet compartments. When either compartment reaches 80% capacity, the system automatically sends an SMS alert to the maintenance staff via a SIM800L GSM module.

An Arduino Uno microcontroller serves as the central processing unit, coordinating all sensor inputs, servo control, and GSM communication. Real-time system output — including waste type detected, servo action, dry and wet bin fill levels, and alert status — is displayed on a connected laptop via the Arduino IDE Serial Monitor at 9600 baud.

This smart garbage interface ensures automated waste segregation, timely collection alerts, reduced manual inspection, improved hygiene, and an eco-friendly environment within gated communities. The system is low-cost, reliable, and suitable for both academic implementation and real-world deployment.

Keywords: Moisture Sensor, Servo Motor, IR Sensor, HC-SR04 Ultrasonic Sensor, Arduino Uno, GSM Module (SIM800L), Waste Segregation, IoT, Smart Waste Management, Dry Waste, Wet Waste.

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CHAPTER 1
INTRODUCTION

In modern urban residential communities, effective waste management is an essential aspect of maintaining hygiene and quality of life. A critical challenge in existing systems is that waste is not segregated at source — dry and wet waste are mixed together, severely complicating downstream recycling and composting. Gated communities typically rely on manual monitoring of garbage bins, which often results in overflowing bins, delayed collection, and unhealthy surroundings for residents.

This project presents the design and development of an Interface of Smart Garbage System for Gated Communities using an Arduino Uno microcontroller. The system incorporates an IR sensor to detect waste at the inlet, a moisture sensor and servo motor for automated waste segregation, two HC-SR04 ultrasonic sensors to independently monitor fill levels of the dry and wet compartments, and a SIM800L GSM module to send automatic SMS alerts. All real-time data is displayed on a connected laptop via the Arduino IDE Serial Monitor.

1.1 AIM

The aim of this project is to design and develop a low-cost, IoT-enabled Smart Garbage System for gated communities that automatically segregates incoming waste into dry and wet categories using a moisture sensor and servo motor, independently monitors the fill levels of both compartments using two separate ultrasonic sensors, uses an IR sensor to trigger each detection cycle, and sends real-time GSM SMS alerts to maintenance staff when either compartment reaches 80% capacity. All system output is displayed on a connected laptop via the Arduino IDE Serial Monitor.

1.2 MOTIVATION

The primary motivation behind this project is the dual problem of unsegregated waste disposal and irregular bin monitoring in gated communities. When wet and dry waste are mixed, recycling and composting become practically impossible, adding to landfill burden. The absence of automated monitoring results in overflowing bins that create hygiene hazards and foul odors for residents.

By deploying an automated segregation and monitoring system, waste is separated at source for better processing, and collection becomes demand-driven based on actual bin levels — reducing unnecessary trips and labor costs. The affordability of the Arduino Uno, moisture sensor, IR sensor, servo motor, HC-SR04 ultrasonic sensors, and SIM800L GSM module makes this system practically implementable as a real-world solution.

1.3 OBJECTIVES

The key objectives of this mini-project are:

1. To implement an IR sensor-triggered detection cycle that automatically initiates the waste classification process when waste is placed at the bin inlet.
2. To automate wet/dry waste segregation using a moisture sensor (threshold: 850) and an SG90 servo motor rotating to 180° for wet waste and 0° for dry waste.
3. To independently monitor the fill levels of the dry waste compartment and wet waste compartment using two separate HC-SR04 ultrasonic sensors (bin height: 13.5 cm).
4. To send automatic GSM SMS alerts to maintenance staff when either compartment exceeds 80% fill level, using the SIM800L module via SoftwareSerial AT commands.

5. To display all real-time output — waste type, servo action, dry fill %, wet fill %, and alert status — on a connected laptop via the Arduino IDE Serial Monitor at 9600 baud.

1.4 PLAN OF ACTION

The project was executed in four phases:

1. **Literature Review and Feasibility Study:** Study of existing smart waste management architectures, IoT-based bin monitoring approaches, waste segregation techniques, and available embedded platforms.
2. **Hardware Design and Prototyping:** Selection and procurement of components (Arduino Uno, FC-28 Moisture Sensor, IR Sensor, SG90 Servo Motor, two HC-SR04 Ultrasonic Sensors, SIM800L GSM Module). Assembly and interfacing on a breadboard with a cardboard bin enclosure divided into dry and wet sections.
3. **Software Development and Integration:** Writing Arduino firmware for IR-triggered detection, moisture sensing, servo control, dual ultrasonic measurement, fill computation, GSM AT command communication, and serial output to laptop. Testing using Arduino IDE Serial Monitor.
4. **Testing, Validation, and Documentation:** System testing under various waste types and fill-level conditions. Verification of segregation accuracy, fill detection, and GSM alert reliability. Documentation of results and conclusions.

CHAPTER 2
LITERATURE REVIEW

The following table summarises key published works on Smart Waste Management Systems, waste segregation techniques, IoT-based bin monitoring, and embedded system approaches reviewed for this project.

S.No.	Author/Source	Year	Method	Key Contribution	Limitation
1	Bharadwaj et al.	2016	Ultrasonic + Arduino	Arduino fill-level detection; proves embedded bin monitoring feasibility	No waste segregation; single compartment only
2	Mahajan et al.	2017	GSM Alert System	SMS alert to sanitation staff when bin full; practical without Wi-Fi	No segregation; no display; SMS cost
3	Sathish et al.	2018	Moisture + Servo	Automated wet/dry segregation using moisture sensing and motorised flap	No fill-level monitoring; no GSM alert
4	Pardini et al.	2019	LoRa + IoT	Long-range low-power bin monitoring using LoRaWAN for large campuses	LoRa gateway needed; costly for small communities
5	Kumar & Patel	2020	Dual Ultrasonic	Independent fill monitoring of two compartments using separate sensors	No automated segregation; manual sorting required
6	Ramson et al.	2021	NB-IoT + Sensor Fusion	Low-power wide-area bin monitoring for smart city waste management	NB-IoT coverage limited; high setup cost
7	Hannan et al.	2021	AI + Computer Vision	Camera-based waste classification using deep learning; enables sorting	High cost; not suitable for low-cost embedded platforms
8	Aazam et al.	2022	Fog Computing + IoT	Edge intelligence reduces latency; fog-layer preprocessing for bin status	High complexity; requires edge computing infrastructure
9	Anagnostopoulos et al.	2022	IoT + Cloud	Cloud-connected bin monitoring with dynamic route optimisation	Requires continuous internet; high infrastructure cost
10	This Work	2026	IR + Moisture + Servo + Dual Ultrasonic + GSM + Arduino	IR-triggered cycle; automated wet/dry segregation; dual fill monitoring; GSM SMS alerts; laptop serial display — all on low-cost Arduino	Single bin prototype; GSM cost per SMS; no cloud dashboard

The literature review confirms that while several works address either waste segregation or fill-level monitoring independently, very few combine IR-triggered detection, automated moisture-based segregation with servo actuation, dual-compartment ultrasonic monitoring, and GSM alerting on a single low-cost Arduino platform. This gap directly motivates the present work.

CHAPTER 3
BLOCK DIAGRAM AND IMPLEMENTATION

The proposed Smart Garbage System adopts a modular design approach combining IR-triggered detection, waste segregation, and dual-compartment bin-level monitoring. This chapter presents the system block diagram and complete workflow.

3.1 SYSTEM BLOCK DIAGRAM

The overall system architecture is illustrated below. The Arduino Uno microcontroller forms the central hub, receiving inputs from four sensors and driving the servo motor and GSM module, while transmitting all data to a connected laptop via USB serial.

IR SENSOR (D6) ──► ARDUINO UNO ──► SERVO MOTOR SG90 (D9)
MOISTURE SENSOR FC-28 (A0) ──► (Central Controller) ──► GSM MODULE SIM800L (D10/D11)
HC-SR04 Dry Bin (D2/D3) ──► ──► LAPTOP Serial Monitor (9600 baud)
HC-SR04 Wet Bin (D4/D5) ──►

Figure 3.1: System Block Diagram of Smart Garbage System

The system consists of four functional blocks: (1) Detection Module — IR sensor (D6) that triggers each waste processing cycle; (2) Segregation Module — FC-28 moisture sensor (A0) for waste classification and SG90 servo motor (D9) for physical direction of waste; (3) Fill-Level Monitoring Module — two HC-SR04 ultrasonic sensors measuring dry (D2/D3) and wet (D4/D5) compartment levels independently; and (4) Output and Communication Module — SIM800L GSM module (D10/D11) for SMS alerts and laptop Serial Monitor for real-time display.

3.2 SYSTEM WORKFLOW

The system workflow proceeds through the following sequential stages, each triggered by IR sensor detection:

1. **IR Sensor Trigger:** The Arduino continuously polls the IR sensor on D6. When an object is detected (LOW output), the system debounces for 80 ms and re-confirms before initiating the processing cycle.
2. **Moisture Sensing and Classification:** The FC-28 moisture sensor on A0 outputs a raw ADC value (0–1023). Values below 850 indicate WET waste; values at or above 850 indicate DRY waste.
3. **Servo Actuation:** The SG90 servo on D9 rotates to 180° (clockwise) for wet waste directing it to the wet bin, or to 0° (counter-clockwise) for dry waste directing it to the dry bin. After 1.5 seconds it returns to the 90° neutral position.
4. **Dual Fill Level Measurement:** Both HC-SR04 sensors are triggered sequentially. Distance = (Echo Duration × 0.0343) / 2 cm. Fill % = ((13.5 − Distance) / 13.5) × 100. Both compartments are measured every cycle.
5. **GSM Alert:** If either compartment fill % ≥ 80, the Arduino sends AT commands to the SIM800L GSM module via SoftwareSerial (D10/D11) to transmit an SMS alert. A flag prevents repeated alerts for the same full-bin event.

6. **Laptop Serial Output:** All data — waste type, servo direction, dry fill %, wet fill %, and alert status — are transmitted to the laptop via USB (9600 baud) and displayed on the Arduino IDE Serial Monitor.

CHAPTER 4
TECHNOLOGIES AND HARDWARE USED

This chapter describes all hardware components, software tools, and libraries used in the development of the Smart Garbage System. Each component's specifications, role, and interfacing details are provided.

4.1 HARDWARE COMPONENTS

S.No.	Component	Specifications and Role
1	Arduino Uno	ATmega328P @ 16 MHz; 32 KB Flash; 2 KB SRAM; 14 digital I/O pins; 6 analog inputs; UART/SPI/I2C. Central controller — reads all sensors, controls servo motor, sends GSM AT commands, and transmits data to laptop via USB serial at 9600 baud.
2	FC-28 Moisture Sensor	Analog output (0–1023); operating voltage 3.3–5 V; sensitivity adjustable via onboard potentiometer. Connected to A0. Classifies incoming waste as Wet (reading < 850) or Dry (reading ≥ 850). Threshold calibrated at 850 in firmware.
3	IR Sensor	Digital output; operating voltage 3.3–5 V; connected to D6. Outputs LOW when an object is detected in front of it. Used to trigger the waste detection cycle automatically when waste is placed at the bin inlet. Debounced in firmware (80 ms).
4	SG90 Servo Motor	Operating voltage: 4.8–6 V; torque: 1.8 kg·cm; PWM signal: 50 Hz. Connected to D9. Rotates to 180° (clockwise) to direct wet waste to the wet bin, and to 0° (counter-clockwise) for dry waste. Returns to 90° neutral after 1.5 seconds.
5	HC-SR04 Ultrasonic Sensor (×2)	Operating voltage: 5 V; frequency: 40 kHz; range: 2–400 cm; resolution: ~3 mm. Sensor 1 (Dry): TRIG D2, ECHO D3. Sensor 2 (Wet): TRIG D4, ECHO D5. Both measure independently every cycle. Bin height: 13.5 cm.
6	SIM800L GSM Module	Quad-band GSM/GPRS; communicates via UART AT commands. SoftwareSerial: RX D10, TX D11. Powered separately at 3.7–4.2 V (peak current ~2 A). Sends SMS alerts to maintenance staff when either bin reaches $\geq 80\%$ fill level.
7	Power Supply	5 V regulated via USB for Arduino Uno, servo, and sensors. Separate 3.7 V Li-ion cell for SIM800L GSM module due to peak current requirement (~2 A during transmission).
8	Laptop (Output Display)	Connected to Arduino Uno via USB. Arduino IDE Serial Monitor at 9600 baud displays all real-time system output: waste type, servo direction, dry fill %, wet fill %, BIN STATUS REPORT, and GSM alert status. Primary monitoring and logging interface.
9	Breadboard & Jumper Wires	Full-size 830-tie-point solderless breadboard for rapid prototyping. Male-to-male and male-to-female DuPont jumper wires for interconnection of all components.

4.2 SOFTWARE TOOLS AND LIBRARIES

S.No.	Tool / Library	Purpose and Usage
1	Arduino IDE (v2.x)	Primary development environment for writing, compiling, and uploading firmware to the Arduino Uno via USB. Serial Monitor (9600 baud) used for real-time output display on the laptop.
2	Servo.h	Official Arduino library for controlling the SG90 servo motor via PWM on D9. Uses sortingServo.write(angle) to set 0° (dry), 90° (neutral), or 180° (wet) positions.
3	SoftwareSerial.h	Enables UART serial communication between the Arduino Uno and the SIM800L GSM module on configurable digital pins (RX: D10, TX: D11). Used to send AT commands for SMS alert generation.

4.3 CIRCUIT IMPLEMENTATION DETAILS

IR Sensor: VCC → 5 V, GND → GND, OUT → Arduino Digital Pin D6. Configured as INPUT in firmware. The main loop() continuously polls digitalRead(D6); LOW triggers the waste processing cycle after 80 ms debounce and re-confirmation.

FC-28 Moisture Sensor: VCC → 5 V, GND → GND, Analog Output (AO) → Arduino Analog Pin A0. The Arduino reads the 10-bit ADC value (0–1023) via analogRead(A0). Values below 850 classify waste as WET; values at or above 850 classify as DRY.

SG90 Servo Motor: VCC → 5 V, GND → GND, Signal → Arduino PWM Pin D9. Controlled via Servo.h library. sortingServo.write(180) rotates to WET bin position; sortingServo.write(0) rotates to DRY bin position; sortingServo.write(90) returns to neutral after 1.5 seconds.

HC-SR04 Sensor 1 (Dry Compartment): VCC → 5 V, GND → GND, TRIG → D2, ECHO → D3. Positioned at the top of the dry bin section. Measures fill level independently for the dry waste compartment every processing cycle.

HC-SR04 Sensor 2 (Wet Compartment): VCC → 5 V, GND → GND, TRIG → D4, ECHO → D5. Positioned at the top of the wet bin section. Measures fill level independently for the wet waste compartment every processing cycle.

SIM800L GSM Module: TX → Arduino D10 (SoftwareSerial RX), RX → Arduino D11 (SoftwareSerial TX). Powered separately from a 3.7 V Li-ion cell. AT commands used: AT (handshake), AT+CMGF=1 (text mode), AT+CMGS (send SMS), Ctrl+Z / ASCII 26 (transmit). Alert flags (alertSentDry, alertSentWet) prevent repeated SMS spam.

Laptop Output: Arduino connected to laptop via USB. Serial.begin(9600) initialises the port. Firmware outputs a formatted BIN STATUS REPORT with Dry and Wet fill levels as ASCII bar charts on the Serial Monitor after every processing cycle.

CHAPTER 5
SOURCE CODE

This chapter presents the complete Arduino firmware for the Smart Garbage Monitoring and Sorting System. The code is written for the Arduino Uno platform and developed using the Arduino IDE. It is structured into clearly commented sections covering pin definitions, system parameters, servo control, ultrasonic fill measurement, GSM alerting, and serial output formatting.

5.1 PROGRAM STRUCTURE

The firmware is organized into the following functional sections:

1. **Section 1 — Pin Definitions:** Maps all hardware components to their Arduino pins (A0, D2–D6, D9–D11).
2. **Section 2 — System Parameters:** Moisture threshold (850), bin height (13.5 cm), and full-alert percentage (80%) defined as adjustable constants.
3. **Section 3 — Servo Positions:** Defines 0° (dry bin), 90° (neutral), and 180° (wet bin) servo angles.
4. **Section 4 — GSM Configuration:** SoftwareSerial on D10/D11 for SIM800L AT command communication.
5. **setup():** Initialises all components, attaches servo to neutral, begins serial ports, and prints startup message.
6. **loop():** Continuously polls IR sensor; on detection triggers the full 10-step processing cycle.
7. **readMoisture():** Reads analog value from A0 and returns raw ADC result (0–1023).
8. **controlServo():** Rotates servo to wet (180°) or dry (0°) position based on waste classification.
9. **getBinFillLevel():** Triggers ultrasonic pulse, measures echo duration, converts to fill percentage using bin height.
10. **sendGSMAlert():** Sends SMS alert via SIM800L using AT commands (AT+CMGF=1, AT+CMGS, Ctrl+Z) when bin reaches threshold.

5.2 COMPLETE SOURCE CODE

The complete firmware listing is provided below. All comments are retained for clarity.

SmartGarbageSystem.ino

```
// ================================================================
//       SMART GARBAGE MONITORING AND SORTING SYSTEM
//                     Arduino UNO
// ================================================================
//  Components:
//    - Moisture Sensor      -> A0
//    - IR Sensor            -> D6
//    - Ultrasonic (Dry Bin) -> TRIG: D2 | ECHO: D3
//    - Ultrasonic (Wet Bin) -> TRIG: D4 | ECHO: D5
//    - Servo Motor          -> D9
//    - GSM Module (SIM800L) -> RX: D10 | TX: D11
// ================================================================

#include <Servo.h>
#include <SoftwareSerial.h>

// ================================================================
```

```
//  SECTION 1: PIN DEFINITIONS
// ================================================================
#define MOISTURE_PIN   A0
#define IR_SENSOR_PIN   6
#define TRIG_DRY        2
#define ECHO_DRY        3
#define TRIG_WET        4
#define ECHO_WET        5
#define SERVO_PIN       9

// ================================================================
//  SECTION 2: ADJUSTABLE SYSTEM PARAMETERS
// ================================================================
const int   MOISTURE_THRESHOLD = 850;   // ABOVE = DRY, BELOW = WET
const float BIN_HEIGHT_CM      = 13.5;  // Physical bin height (cm)
const int   FULL_ALERT_PERCENT = 80;    // GSM alert threshold (%)

// ================================================================
//  SECTION 3: SERVO ANGLE POSITIONS
// ================================================================
const int SERVO_NEUTRAL  = 90;   // Resting / center position
const int SERVO_WET_BIN  = 180;  // Clockwise       -> WET bin
const int SERVO_DRY_BIN  = 0;    // Counter-clockwise -> DRY bin

// ================================================================
//  SECTION 4: GSM CONFIGURATION
// ================================================================
SoftwareSerial gsmSerial(10, 11);
const String PHONE_NUMBER = "+918121847746";

// ================================================================
//  SECTION 5: GLOBAL OBJECTS & VARIABLES
// ================================================================
Servo sortingServo;
bool  alertSentDry = false;
bool  alertSentWet = false;

// ================================================================
//  SETUP
// ================================================================
void setup() {
  Serial.begin(9600);
  sortingServo.attach(SERVO_PIN);
  sortingServo.write(SERVO_NEUTRAL);
  pinMode(TRIG_DRY, OUTPUT);  pinMode(ECHO_DRY, INPUT);
  pinMode(TRIG_WET, OUTPUT);  pinMode(ECHO_WET, INPUT);
  pinMode(IR_SENSOR_PIN, INPUT);
  gsmSerial.begin(9600);
  delay(500);
  Serial.println(F("================================================"));
  Serial.println(F("   Smart Garbage Monitoring & Sorting System    "));
  Serial.println(F("================================================"));
  Serial.println(F(" Moisture Threshold : 850"));
  Serial.println(F(" Bin Height         : 13.5 cm"));
  Serial.println(F(" Trigger Mode       : IR Sensor (Automatic)"));
  Serial.println(F("================================================"));
  Serial.println(F(" System Ready. Waiting for trash..."));
  delay(2000);
}
```

```
// ================================================================
//  MAIN LOOP
// ================================================================
void loop() {
  if (digitalRead(IR_SENSOR_PIN) == LOW) {
    delay(80);
    if (digitalRead(IR_SENSOR_PIN) == LOW) {
      printSectionHeader("NEW CYCLE STARTING");
      Serial.println(F("  [IR SENSOR] >>> Trash Detected!"));
      delay(300);
      int moistureValue = readMoisture();
      bool isWet = (moistureValue < MOISTURE_THRESHOLD);
      Serial.print(F("  Moisture Sensor Value : "));
      Serial.println(moistureValue);
      Serial.print(F("  Waste Type Detected   : "));
      if (isWet) Serial.println(F("*** WET WASTE ***"));
      else        Serial.println(F("*** DRY WASTE ***"));
      controlServo(isWet);
      delay(1500);
      Serial.println(F("  Measuring bin fill levels..."));
      delay(300);
      float dryFillPercent = getBinFillLevel(TRIG_DRY, ECHO_DRY);
      float wetFillPercent = getBinFillLevel(TRIG_WET, ECHO_WET);
      Serial.println(F("  ----------------------------------------"));
      Serial.println(F("  |         BIN STATUS REPORT            |"));
      Serial.println(F("  ----------------------------------------"));
      Serial.print(F("  | Dry Bin Fill Level  : "));
      printPaddedPercent(dryFillPercent);
      Serial.print(F("  | Wet Bin Fill Level  : "));
      printPaddedPercent(wetFillPercent);
      Serial.println(F("  ----------------------------------------"));
      if (dryFillPercent >= FULL_ALERT_PERCENT) {
        Serial.println(F("  ! WARNING: DRY BIN IS FULL!"));
        if (!alertSentDry) { sendGSMAlert("DRY"); alertSentDry = true; }
      } else { alertSentDry = false; }
      if (wetFillPercent >= FULL_ALERT_PERCENT) {
        Serial.println(F("  ! WARNING: WET BIN IS FULL!"));
        if (!alertSentWet) { sendGSMAlert("WET"); alertSentWet = true; }
      } else { alertSentWet = false; }
      sortingServo.write(SERVO_NEUTRAL);
      Serial.println(F("  Cycle complete. Waiting for next trash..."));
      printSectionFooter();
      while (digitalRead(IR_SENSOR_PIN) == LOW) delay(50);
      delay(500);
    }
  }
}

// ================================================================
//  FUNCTION: readMoisture()
// ================================================================
int readMoisture() {
  return analogRead(MOISTURE_PIN);
}

// ================================================================
//  FUNCTION: controlServo(bool isWet)
// ================================================================
void controlServo(bool isWet) {
  if (isWet) {
```

```
    Serial.println(F("  Servo Action : Rotating -> WET BIN  (180 clockwise)"));
    sortingServo.write(SERVO_WET_BIN);
  } else {
    Serial.println(F("  Servo Action : Rotating -> DRY BIN  (0 counter-CW)"));
    sortingServo.write(SERVO_DRY_BIN);
  }
}

// ================================================================
//  FUNCTION: getBinFillLevel(int trigPin, int echoPin)
//  Sends ultrasonic pulse, measures echo, returns fill %
// ================================================================
float getBinFillLevel(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH, 30000);
  float distance = (duration * 0.0343) / 2.0;
  if (duration == 0) distance = BIN_HEIGHT_CM;
  distance = constrain(distance, 0.0, BIN_HEIGHT_CM);
  float fillPercent = ((BIN_HEIGHT_CM - distance) / BIN_HEIGHT_CM) * 100.0;
  return constrain(fillPercent, 0.0, 100.0);
}

// ================================================================
//  FUNCTION: sendGSMAlert(String binType)
//  Sends SMS via SIM800L using AT commands
// ================================================================
void sendGSMAlert(String binType) {
  String msg = "ALERT: " + binType +
               " bin is FULL! Please empty it. - SmartGarbage System";
  Serial.println(F("  [GSM] Preparing alert SMS..."));
  Serial.print(F("  [GSM] To      : ")); Serial.println("+918121847746");
  Serial.print(F("  [GSM] Message : ")); Serial.println(msg);
  gsmSerial.println("AT");            delay(1000);
  gsmSerial.println("AT+CMGF=1");    delay(1000);
  gsmSerial.print("AT+CMGS=\"");
  gsmSerial.print("+918121847746");
  gsmSerial.println("\"");            delay(1000);
  gsmSerial.print(msg);               delay(500);
  gsmSerial.write(26);                delay(5000);
  Serial.println(F("  [GSM] Status  : Message Sent Successfully!"));
}

// ================================================================
//  HELPERS: Serial Monitor formatting
// ================================================================
void printSectionHeader(const char* title) {
  Serial.println(F("================================================"));
  Serial.print(F("  ")); Serial.println(title);
  Serial.println(F("================================================"));
}
void printSectionFooter() {
  Serial.println(F("================================================"));
}
void printPaddedPercent(float percent) {
  Serial.print(F("  |   ")); Serial.print(percent, 1); Serial.print(F("%   ["));
  int filled = (int)(percent / 10);
```

```
  for (int i = 0; i < 10; i++) {
    if (i < filled) Serial.print(F("#"));
    else            Serial.print(F("-"));
  }
  Serial.println(F("]"));
}
// ================================================================
//  END OF PROGRAM
// ================================================================
```

CHAPTER 6
RESULTS AND OBSERVATIONS

The Smart Garbage System was tested using a prototype bin constructed from a cardboard enclosure divided into two compartments — a dry waste section and a wet waste section — with the servo motor mounted at the inlet to direct waste. The system was evaluated for IR trigger accuracy, waste segregation correctness, independent fill-level detection, GSM alert delivery, and Serial Monitor output. All results are presented with actual photographs and Serial Monitor screenshots captured during testing.

5.1 SYSTEM SETUP — FULL CIRCUIT

Figure 5.1 shows the complete hardware prototype assembled on a breadboard. The cardboard box serves as the bin enclosure with a servo-controlled divider flap at the top centre. The two HC-SR04 ultrasonic sensors are mounted on the left and right sides of the bin to monitor the dry and wet compartments respectively. The moisture sensor is positioned at the inlet. The Arduino Uno, SIM800L GSM module (powered by a Li-ion battery), and breadboard circuit are placed in front of the bin. The laptop connected via USB serves as the output display.

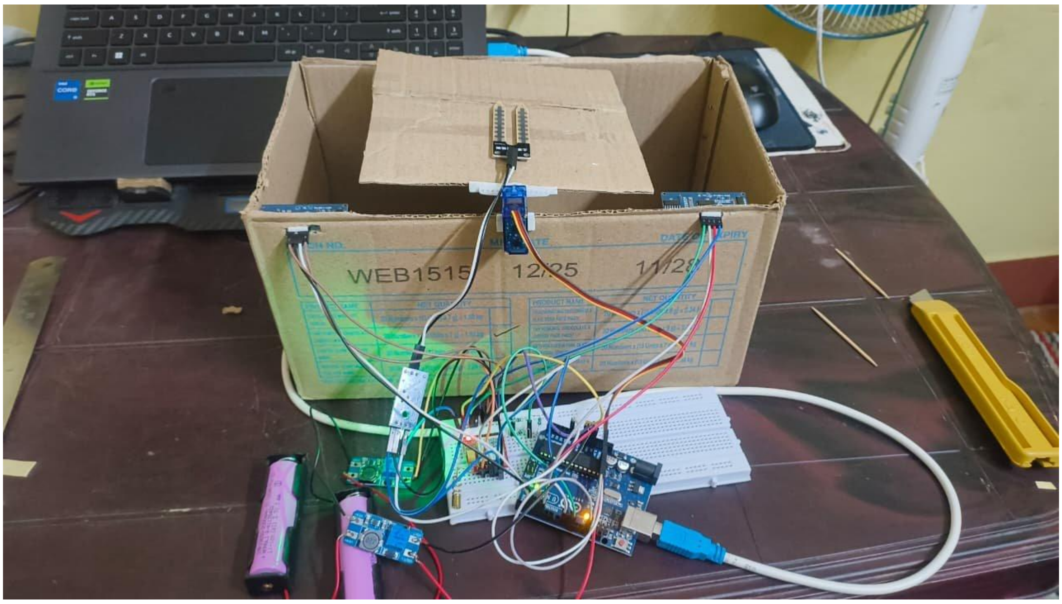

Figure 5.1: Figure 5.1: Complete Hardware Prototype — Cardboard Bin with Servo, Ultrasonic Sensors, Moisture Sensor, Arduino Uno, GSM Module, and Laptop

5.2 WET WASTE DETECTION AND SEGREGATION

Figure 5.2 shows the Serial Monitor output and bin photograph when wet waste was detected. The moisture sensor reading fell below the threshold of 850, causing the system to classify the waste as WET WASTE. The servo motor rotated 180° clockwise to direct the waste into the wet bin compartment. The BIN STATUS REPORT displayed Dry Bin Fill Level: 0% and Wet Bin Fill Level: 4.6%, confirming the ultrasonic sensors correctly measured both compartments independently after the waste entered the wet section.

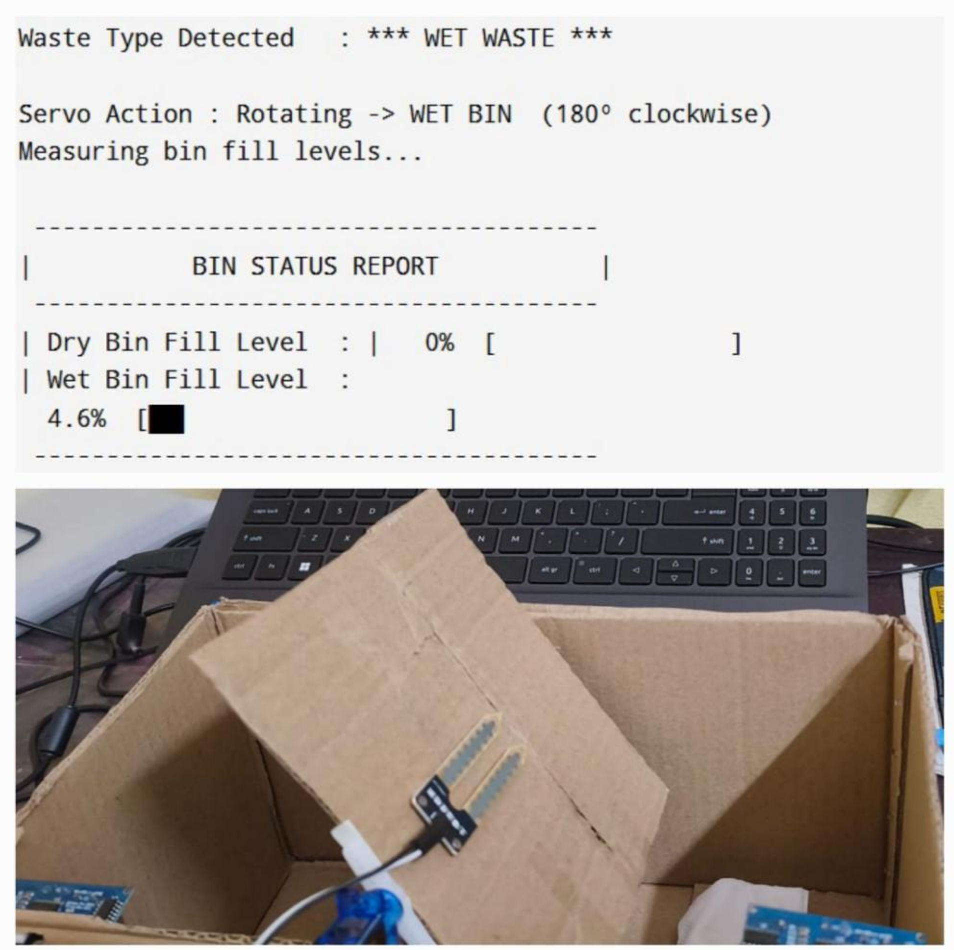

Figure 5.2: Figure 5.2: Wet Waste Detected — Serial Monitor Output (WET WASTE, Servo 180° clockwise) and Bin Photograph

5.3 DRY WASTE DETECTION AND SEGREGATION

Figure 5.3 shows the Serial Monitor output and bin photograph when dry waste was detected. The moisture sensor reading was at or above the threshold of 850, causing the system to classify the waste as DRY WASTE. The servo motor rotated 0° counter-clockwise to direct the waste into the dry bin compartment. The BIN STATUS REPORT displayed Dry Bin Fill Level: 8.5% and Wet Bin Fill Level: 0.0%, confirming accurate independent fill-level measurement by both ultrasonic sensors.

```
Waste Type Detected   : *** DRY WASTE ***

Servo Action : Rotating -> WET BIN  (180° clockwise)
Measuring bin fill levels...

 --------------------------------------
|           BIN STATUS REPORT          |
 --------------------------------------
| Dry Bin Fill Level  : |   8.5%  [█                   ]
| Wet Bin Fill Level  :
  0.0%  [                    ]
```

Figure 5.3: Figure 5.3: Dry Waste Detected — Serial Monitor Output (DRY WASTE, Servo 0° counter-clockwise) and Bin Photograph

5.4 BIN FULL — GSM ALERT NOTIFICATION

Figure 5.4 shows the Serial Monitor output and bin photograph when the wet bin reached 80.6% fill level, triggering the GSM alert. The system printed a WARNING: DRY BIN IS FULL message and immediately initiated the GSM SMS sequence: the SIM800L module was addressed via AT commands, the alert message 'ALERT: DRY bin is FULL! Please empty it immediately. - SmartGarbage System' was composed, and the SMS was sent to +91 81218 47746. The Serial Monitor confirmed '[GSM] Status: Message Sent Successfully!' The alert flag was set to prevent repeated SMS for the same event.

```
| Wet Bin Fill Level  : |  80.6%  [████████████████   ]
| Dry Bin Fill Level  :
  7.4%  [█                  ]
 ---------------------------------------

! WARNING: DRY BIN IS FULL!
--------------------------------------------------
[GSM] Preparing alert SMS...
[GSM] To      : +91 81218 47746
[GSM] Message : ALERT: DRY bin is FULL! Please empty it immediately.
    SmartGarbage System
[GSM] Status  :  Message Sent Successfully!
--------------------------------------------------
```

Figure 5.4: Figure 5.4: Bin Full Alert — Serial Monitor (80.6% Wet Bin, GSM SMS Sent) and Bin Photograph

5.5 COMPARISON: CONVENTIONAL VS SMART GARBAGE SYSTEM

Parameter	Conventional System	Smart Garbage System (This Work)
Waste Detection Trigger	Manual — by residents	Automatic — IR sensor (D6)
Waste Segregation	Manual sorting	Automated — moisture sensor + SG90 servo
Dry Bin Monitoring	Manual inspection	HC-SR04 Sensor 1 (D2/D3) — independent
Wet Bin Monitoring	Not tracked separately	HC-SR04 Sensor 2 (D4/D5) — independent
Alert Mechanism	None / delayed manual report	Automatic GSM SMS via SIM800L
Real-Time Display	None	Laptop Serial Monitor (9600 baud)
Segregation Accuracy	Inconsistent (human error)	Consistent — moisture threshold 850
Response Time	Hours to days	Within 1 cycle (~1-2 sec)
Cost	High labor cost	Low-cost embedded hardware

Table 5.1: Conventional vs Smart Garbage System — Feature Comparison

The comparison clearly demonstrates that the Smart Garbage System provides automated waste detection, segregation, dual-compartment fill monitoring, and GSM-based alerts — significantly outperforming conventional manual systems in all key parameters.

CHAPTER 7
CONCLUSION

This mini-project successfully demonstrated the design and development of an IoT-based Smart Garbage System for gated communities using an Arduino Uno microcontroller. The system integrates an IR sensor for automatic waste detection, an FC-28 moisture sensor for waste type classification, an SG90 servo motor for physical waste segregation into dry and wet compartments, two HC-SR04 ultrasonic sensors for independent fill-level monitoring of each compartment, and a SIM800L GSM module for automatic SMS alert notifications to maintenance staff.

The IR sensor reliably triggered the detection cycle whenever waste was placed at the bin inlet, eliminating the need for manual initiation. The moisture sensor correctly classified waste as wet or dry based on a calibrated threshold of 850, and the servo motor responded by rotating to the appropriate bin position (180° for wet, 0° for dry). Both ultrasonic sensors independently measured fill levels with good accuracy, and the system correctly triggered GSM SMS alerts when either bin reached 80% capacity, with the message 'ALERT: [BIN TYPE] bin is FULL! Please empty it immediately.' confirmed as sent on the Serial Monitor.

The laptop Serial Monitor provided clear, real-time formatted output — including waste type, servo action, BIN STATUS REPORT with ASCII fill bars for both compartments, and GSM alert status — serving as an effective monitoring and logging interface without requiring any additional display hardware.

In conclusion, the integration of IR-triggered detection, moisture-based waste segregation, dual ultrasonic fill monitoring, and GSM alerting on a single low-cost Arduino platform is technically feasible and delivers significant improvements over conventional manual waste management systems. This work establishes a strong foundation for practical deployment of smart waste management in gated communities.

CHAPTER 8
FUTURE SCOPE

The current implementation provides a strong proof-of-concept. The following enhancements are proposed for future development:

- **Wi-Fi / Cloud Dashboard:** Replace USB-to-laptop serial with Wi-Fi (ESP8266/ESP32) to transmit bin data to cloud platforms (ThingSpeak, Firebase) for remote monitoring via web or mobile dashboard.
- **Multi-Bin Network:** Extend the system to manage multiple smart bins distributed across the gated community, with a centralised dashboard showing fill levels of all bins simultaneously.
- **Additional Waste Categories:** Add more sensors (gas sensor for organic detection, weight sensor) to classify waste into more categories beyond binary wet/dry for better waste processing.
- **AI-Based Fill Rate Prediction:** Train a lightweight machine learning model on historical fill rate data to predict when a bin will reach capacity, enabling proactive scheduling of waste collection.
- **Solar-Powered Operation:** Integrate a solar panel and rechargeable battery for autonomous off-grid operation, eliminating dependency on mains power for outdoor deployment.
- **Automated Bin Compaction:** Add a DC motor-driven compaction mechanism that compresses waste when the bin reaches a defined fill level, increasing capacity and reducing collection frequency.
- **Custom PCB and Weatherproof Enclosure:** Design a custom PCB to replace the breadboard prototype and fabricate a weatherproof enclosure for robust, deployable outdoor installation.

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SUSTAINABLE DEVELOPMENT GOALS (SDG)

This project — Interface of Smart Garbage System in Gated Communities — directly contributes to two United Nations Sustainable Development Goals through its application of IoT-based embedded technology for intelligent waste management.

SDG 11: Sustainable Cities and Communities

Target 11.6 — Reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal waste management.

How this project contributes: Gated communities are a fundamental unit of urban living. This system automates waste segregation at source, ensuring that dry and wet waste are separated before they enter the collection stream — directly reducing the burden on municipal solid waste infrastructure. The dual ultrasonic sensors enable proactive bin monitoring so that collection is triggered precisely when needed, preventing overflow, foul odors, and unhygienic conditions that degrade urban living quality. By replacing reactive, manual waste collection with an intelligent, IoT-driven system, this project directly advances the goal of making cities cleaner, healthier, and more sustainable.

SDG 12: Responsible Consumption and Production

Target 12.5 — By 2030, substantially reduce waste generation through prevention, reduction, recycling, and reuse.

How this project contributes: The single most impactful action in improving waste processing efficiency is segregation at source. By automatically classifying waste as wet or dry using a moisture sensor and physically directing it using a servo motor, this system ensures that recyclable dry waste (paper, plastic, metal) is never contaminated by wet organic waste. This dramatically improves the quality and quantity of material that can be recycled or composted, directly supporting responsible production and consumption cycles. The GSM alert mechanism ensures bins are emptied before overflow occurs, further reducing waste spillage and environmental contamination. By making waste segregation effortless and automatic for residents, the system promotes responsible consumption habits without requiring behavioral change.

SDG	Target	Project Contribution
SDG 11: Sustainable Cities and Communities	11.6 — Municipal waste management	Automated bin monitoring prevents overflow; proactive GSM alerts enable timely collection; improves urban hygiene in gated communities
SDG 12: Responsible Consumption and Production	12.5 — Reduce waste through recycling and reuse	Automatic wet/dry segregation at source protects recyclable dry waste from contamination; maximises recycling potential and reduces landfill contribution

Table: SDG Alignment of Smart Garbage System

By combining IoT-enabled bin monitoring, automated waste segregation, and real-time alert communication in a single low-cost embedded system, this project demonstrates that sustainable waste management is achievable without large-scale infrastructure investment — making it applicable to gated communities, educational institutions, commercial complexes, and beyond.